Kubernetes Events and Log Ingestion

> **Series:** K8S | **Notebook:** 7 of 9 | **Created:** January 2026

Capturing and Analyzing Kubernetes Events and Logs

Kubernetes events and container logs provide crucial insights for debugging and operational awareness. This notebook covers event monitoring, log ingestion configuration, and analysis patterns in Dynatrace.

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Prerequisites

Requirement	Details
Dynatrace Environment	SaaS with log ingestion enabled
DynaKube	ActiveGate with `kubernetes-monitoring`
Permissions	`logs.read`, `logs.ingest`, `events.read`
Knowledge	K8S-01 Fundamentals

1. Kubernetes Events Overview

Event Types

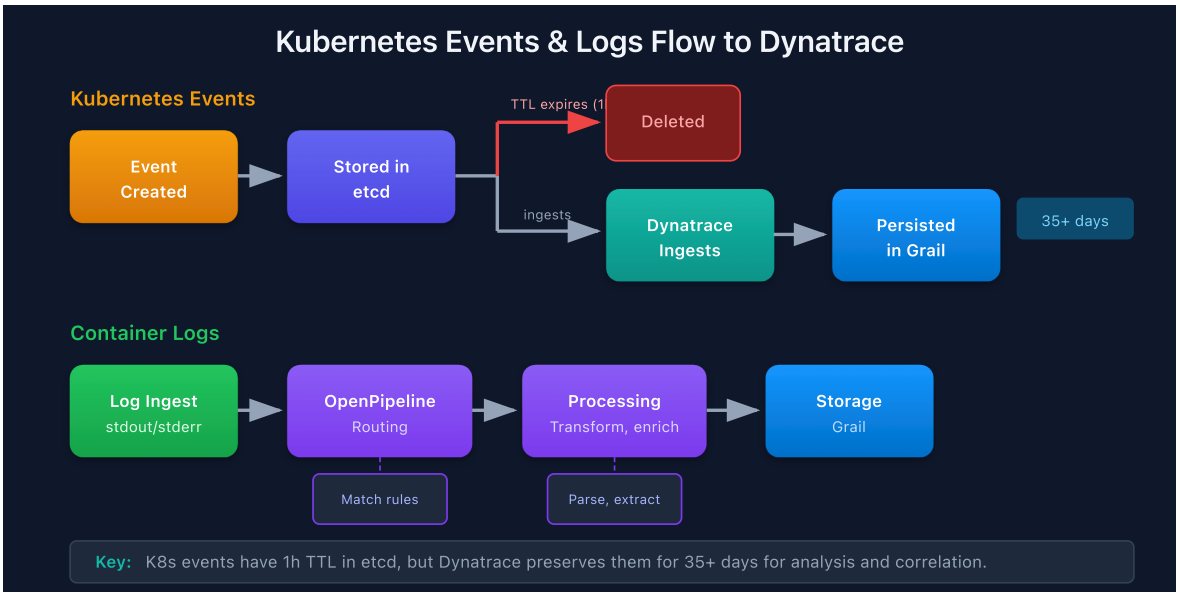
Kubernetes events are first-class objects that record what happened in the cluster.

Type	Description	Examples
Normal	Routine operations	Scheduled, Pulled, Created
Warning	Potential issues	FailedScheduling, BackOff, Unhealthy

Event Sources

Source	Events Generated
Scheduler	Scheduling decisions, failures
Kubelet	Pod lifecycle, probe results
Controller Manager	ReplicaSet scaling, Deployment rollouts
Custom Controllers	CRD reconciliation events

Event Lifecycle



Important: Kubernetes events are short-lived. Dynatrace captures them for long-term storage and analysis.

2. Event Ingestion Configuration

ActiveGate Kubernetes Monitoring

Enable event collection via DynaKube:

```
spec:
  activeGate:
    capabilities:
      - kubernetes-monitoring  # Enables event collection
      - routing
```

Event Filtering

Configure which events to ingest:

Setting	Location	Purpose
Event types	Settings > Cloud and virtualization	Normal, Warning, or both
Namespace filter	DynaKube spec	Limit to specific namespaces
Event reasons	Settings	Include/exclude specific reasons

Recommended Configuration

Use Case	Configuration
Full visibility	All event types, all namespaces
Warning focus	Warning events only, all namespaces
Cost optimization	Warning events, specific namespaces

```
// Recent Kubernetes events
fetch logs
| filter matchesPhrase(log.source, "kubernetes") or matchesPhrase(log.source,
"k8s")
| fields timestamp, content
| sort timestamp desc
| limit 50
```

```
// Warning events only
fetch logs
| filter matchesPhrase(content, "Warning")
| filter matchesPhrase(log.source, "kubernetes") or matchesPhrase(log.source,
"k8s")
| fields timestamp, content
| sort timestamp desc
| limit 30
```

3. Container Log Collection

Log Collection Methods

Method	Source	Configuration
OneAgent	Container stdout/stderr	Automatic with OneAgent
Log Monitoring	Mounted log files	Custom paths in settings
Fluentd/Fluent Bit	External forwarder	OTLP endpoint

OneAgent Log Collection

OneAgent automatically collects:
- Container stdout logs
- Container stderr logs
- Process-generated log files (with configuration)

Log Attributes

Attribute	Source	Example
`k8s.namespace.name`	Container metadata	`checkout`
`k8s.pod.name`	Container metadata	`checkout-api-abc123`
`k8s.container.name`	Container metadata	`api`
`dt.entity.container_group_instance`	Entity relationship	`CONTAINER_GROUP_INSTANCE-XXX`
`loglevel`	Parsed from content	`ERROR`, `WARN`, `INFO`

DynaKube Log Configuration

```
spec:
  oneAgent:
    cloudNativeFullStack:
      env:
        - name: ONEAGENT_ENABLE_LOG_ANALYTICS
          value: "true"
```

```
// Container logs by namespace
fetch logs
| filter isNotNull(k8s.namespace.name)
| summarize logCount = count(), by:{k8s.namespace.name}
| sort logCount desc
| limit 15
```

```
// Error logs with Kubernetes context
fetch logs
| filter loglevel == "ERROR" or loglevel == "SEVERE"
| filter isNotNull(k8s.namespace.name)
| fields timestamp, k8s.namespace.name, k8s.pod.name, content
| sort timestamp desc
| limit 30
```

4. OpenPipeline for K8s Logs

Log Processing Pipeline

OpenPipeline processes logs before storage:

```
Log Ingest → Routing → Processing → Storage
               ↓           ↓
         Match rules   Transform, enrich
```

Common Processing Rules

Rule Type	Use Case	Example
Parse	Extract fields	JSON parsing, regex
Transform	Modify content	Rename fields, mask data
Filter	Drop logs	Remove debug logs
Route	Direct to bucket	By namespace or app

Example: Parse JSON Logs

```
# OpenPipeline configuration
pipelines:
  - name: k8s-json-logs
    routes:
      - match: k8s.namespace.name exists
    processing:
      - type: json
        source: content
```

Example: Filter Debug Logs

```
pipelines:
  - name: k8s-filter-debug
    routes:
      - match: k8s.namespace.name exists and loglevel == "DEBUG"
    processing:
      - type: drop
```

5. Event Analysis Patterns

Key Event Reasons to Monitor

Reason	Meaning	Action
FailedScheduling	Pod can't be scheduled	Check resource availability
FailedMount	Volume mount failed	Check PV/PVC config
BackOff	Container restart backoff	Check logs, fix crash
Unhealthy	Probe failed	Check probe config, app health
Evicted	Pod evicted from node	Check node pressure
OOMKilled	Out of memory	Increase limits

```
// Failed scheduling events
fetch logs
| filter matchesPhrase(content, "FailedScheduling")
| fields timestamp, content
| sort timestamp desc
| limit 20
```

```
// Pod restart events (BackOff, CrashLoopBackOff)
fetch logs
| filter matchesPhrase(content, "BackOff") or matchesPhrase(content,
"CrashLoopBackOff")
| fields timestamp, content
| sort timestamp desc
| limit 20
```

```
// Volume mount failures
fetch logs
| filter matchesPhrase(content, "FailedMount") or matchesPhrase(content,
"FailedAttachVolume")
| fields timestamp, content
| sort timestamp desc
| limit 20
```

```
// Event frequency by reason (last 24h)
fetch logs, from: now() - 24h
| filter matchesPhrase(log.source, "kubernetes") or matchesPhrase(log.source,
"k8s")
| filter matchesPhrase(content, "Warning")
| summarize eventCount = count(), by:{bin(timestamp, 1h)}
| sort timestamp asc
```

6. Log Analysis Patterns

Error Log Investigation

```
// Pattern: Find errors with full context
fetch logs
| filter loglevel == "ERROR"
| filter k8s.namespace.name == "checkout"
| fields timestamp, k8s.pod.name, content
| sort timestamp desc
| limit 50
```

Log Volume Analysis

```
// Pattern: Identify noisy pods
fetch logs, from: now() - 1h
| summarize count(), by:{k8s.pod.name}
| sort count() desc
| limit 10
```

Exception Tracking

```
// Pattern: Find stack traces
fetch logs
| filter matchesPhrase(content, "Exception") or matchesPhrase(content,
"Traceback")
| fields timestamp, k8s.namespace.name, content
| sort timestamp desc
```

```
// Log volume by pod (find noisy pods)
fetch logs, from: now() - 1h
| filter isNotNull(k8s.pod.name)
| summarize logCount = count(), by:{k8s.pod.name}
| sort logCount desc
| limit 15
```

```
// Exception and error messages
fetch logs
| filter matchesPhrase(content, "Exception") or matchesPhrase(content,
"error") or matchesPhrase(content, "failed")
| filter isNotNull(k8s.namespace.name)
| fields timestamp, k8s.namespace.name, k8s.pod.name, content
| sort timestamp desc
| limit 30
```

```
// Log level distribution by namespace
fetch logs, from: now() - 1h
| filter isNotNull(k8s.namespace.name) and isNotNull(loglevel)
| summarize count(), by:{k8s.namespace.name, loglevel}
| sort count() desc
| limit 30
```

7. Alerting on Events and Logs

Event-Based Alerts

Alert	Condition	Severity
Failed Scheduling	FailedScheduling events > 5 in 10 min	Warning
Crash Loop	CrashLoopBackOff events	Warning
OOM Kills	OOMKilled events	Critical
Volume Failures	FailedMount events	Critical

Log-Based Alerts

Alert	Condition	Severity
Error Spike	Error log count > baseline	Warning
Critical Errors	Specific error patterns	Critical
No Logs	Log volume drops to 0	Warning

Custom Metric Events

Create metric events for log-based alerting:

1. Navigate to **Settings > Anomaly detection > Custom events**
2. Create event based on log count metric
3. Set thresholds and notification targets

Next Steps

With event and log monitoring configured, proceed to:

Next Notebook	Topic
K8S-08: DQL for Kubernetes	Advanced query patterns
K8S-09: Troubleshooting	Debugging K8s monitoring

Summary

In this notebook, you learned:

- Kubernetes event types and sources
- Event ingestion configuration via DynaKube
- Container log collection methods
- OpenPipeline for log processing
- Event analysis patterns for common issues

- Log analysis patterns for debugging
- Alerting strategies for events and logs

References

- Kubernetes Log Monitoring
- Kubernetes Events
- OpenPipeline
